MIL-C-46239B(MU)
15 May 1974
SUPERSEDING
MIL-C-46239A(MU)
29 March 1968

MILITARY SPECIFICATION
CARTRIDGE, IMPULSE, M31A2 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Impulse, M31A2, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS 28741	- Hose, Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Armament Command

D11731602	- Cartridge, Impulse, M31A2, Assembly
C9258147	- Box, Packing, Ammunition for Cartridge, Impulse, M31A2

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

C9258146	- Box, Fiberboard (Inner) Packing, Ammunition for Cartridge Impulse, M31A2
C9258148	- Box, Fiberboard (Outer) Alternative, Packing, Ammunition for Cartridge, Impulse, M31A2
IEL 11731602	- Inspection Equipment List for Cartridge, Impulse, M31A2, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Vibration.-The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.3 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.4 Functional.

3.4.1 The cartridge shall not fail to fire.

3.4.2 Ballistics.-The cartridge when assembled and tested in an acceptable M3A1 Canopy Remover shall propel a 300 pound mass vertically and shall comply with the requirements of Table I.

TABLE I

Temperature Degrees (°) Fahrenheit (F)	Minimum Velocity Feet Per Second (ft/sec)	Thrust Pounds Minimum (Min.)	Ignition Delay Milliseconds (See 6.1)
-65° to -70°	20	2000	40 Max.
65° to 75°	20.5	2600	40 Max.
160° to 165°	22.5	3200	40 Max.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

3.4.2.1 Initiation.-The remover shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of Aircraft Hose, Size No. 4, in accordance with MS 28741, at $70^{\circ} \pm 5^{\circ}$F.

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, specifications and the following:

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 cartridges for sealant test, 21 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistics sample) and 10 sets of unassembled components for dimensional and visual examination and tests.

The sample shall be manufactured using the same materials, equipment, processes and procedures as sued in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.2.1.1 Examination and tests.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor class, the criteria in Table III shall be used. All non-conforming material shall be rejected. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. The roughness comparison standards prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

TABLE III

	Total Allowable Defects Per Sample Size			
Sample Size for Major Defects - - - - - - -	110	150	225	300
No. of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects - - - - - - -	110	150	225	300
No. of Charactersitics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Firing Plug, Drawing B8593498, covering a detail of Drawing D11731602.

Defect	Method of Inspection
CRITICAL	
1. Length of striker tip, min.	Gage
MAJOR	
101. Diameter of body, min.	Gage
102. Profile of striker tip	Gage
103. Height of boss	Gage
104. Metal defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of boss	Gage
203. Total length	Gage
204. Length of striker tip, max.	Gage
205. Surface finish improper	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.2 Retaining Ring, Drawing B8593499, covering a detail of Dwg. D11731602.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of center hole with body	Gage
102. Diameter of body, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of center hole	Gage
202. Thickness of body, max.	Gage

4.3.2.3 Cartridge Case, Drawing C11731600, covering a detail of Dwg. D11731602.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of body, min.	Gage
104. Thickness of base, min.	Gage
105. Thickness of sidewall, min.	Gage
106. Diameter of head, min.	Gage
107. Metal defective	Visual
108. Protective finish inadequate or defective	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of head, max.	Gage
203. Total length	Gage
204. Thickness of sidewall, max.	Gage
205. Location of datum diameter	Gage
206. Radius or chamfer missing or incorrect	Visual
207. Surface finish improper	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.4 Sealing Disc, Drawing C8593494, covering a detail of Dwg. D11731602.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Marking missing, incorrect or illegible	Visual
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage
203. Varnish missing	Visual

4.3.2.5 Igniting Disc, Drawing B8593355, covering a detail of Dwg. D11731602.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage

4.3.2.6 Head, Drawing C11731601, covering a detail of Drawing D11731602.

Defect	Method of Inspection
CRITICAL	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of flange with body	Gage
102. Concentricity of counterbore with firing plug cavity	Gage
103. Concentricity of firing plug cavity with body	Gage
104. Concentricity of firing plug cavity with primer percussion cavity	Gage
105. Diameter of counterbore at flange end, max.	Gage
106. Wall thickness, min.	Gage
107. Surface finish firing plug cavity primer percussion cavity, and face of flange	Visual
108. Diameter of firing plug cavity	Gage
109. Diameter of percussion primer large cavity	Gage
110. Diameter of percussion primer small cavity	Gage
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual
113. Burr	Visual
114. Foreign matter	Visual
MINOR	
201. Diameter of counterbore at flange end, min.	Gage
202. Total length	Gage
203. Depth of counterbore at flange end	Gage
204. Depth from flange face to bottom of firing plug cavity	Gage
205. Depth from flange face to bottom of large percussion primer cavity	Gage
206. Diameter of body	Gage
207. Diameter at rear	Gage
208. Width of undercut, rear of percussion primer small cavity	Gage
209. Depth from rear face to bottom of igniter cavity	Gage
210. Width of undercut, rear of igniter cavity	Gage
211. Depth from rear face to start of igniter cavity	Gage
212. Width of spring retaining cavity	Gage
213. Width of flange	Gage
214. Diameter of spring retaining cavity	Gage
215. Chamfers and radius missing or incorrect	Visual
216. Surface finish improper except as otherwise specified	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.7 Retaining Spring, Dwg. B8593742, covering a detail of Dwg. D11731602.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Thickness of wire, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, free	Gage
202. Thickness of wire, max.	Gage
203. Gap, min. (when spring is compressed to specified O.D.)	Gage

4.3.2.8 Primer, Percussion, Dwg. D10551686, covering a detail of Dwg. D11731602.

CRITICAL	
1. Primer foil discolored	Visual
2. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.9 Cartridge, Impulse, M31A2, Assembly, Drawing D11731602.

CRITICAL	
1. Improper assembly	X-ray 1/ - Visual
2. Any part missing	X-ray 1/ - Visual
3. Primer pellet broken, cracked or dislodged	X-ray 1/ - Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.9 (Cont'd)

Defect	Method of Inspection
4. Metal defective	X-ray 1/ - Visual
5. Primer not properly seated	X-ray 1/ - Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Crimp missing or improper	Visual
103. "O" - ring cut, split or cracked	Visual
104. Sealant missing	Visual
105. Sealant in cavity	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of head, max.	Gage
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Varnish missing or inadequate	Visual
206. Fold, wrinkle or bulge	Visual

1/ The cartridge shall be x-rayed with the sides of the case perpendicular to the rays. The x-rays negatives properly identified shall be forwarded to the procuring activity.

4.3.2.10 Box, Fiberboard (Inner), Packing, Ammunition for Cartridge, Impulse, M31A2, Drawing C9258146.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated layers	Visual
206. Marking missing, incorrect or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.11 Box, Packing, Ammunition for Cartridge, Impulse, M31A2, Dwg. C9258147.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardward missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.2.12 Box, Fiberboard (Outer) Alternative, Drawing C9258148.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Box broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.3.3.1 Vibration.-Twelve (12) of the twenty-one (21) x-rayed cartridges of the initial production sample, selected for ballistic tests and properly identified, shall be subjected to vibration as prescribed in 3.2. The twelve (12) samples for vibration shall be grouped and temperature conditioned, prior to vibration, as indicated in Table IV below. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.9. Evidence of any disintegration of the primer, propellant, ignitor, or derangement of any component shall be cause for rejection initial production sample. When a change occurs in the lot of primers, propellant, or igniters, after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph.

4.3.3.2 Airtightness.-Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.3 Hardness.-Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.4 Protective finish.-Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.5 Sealant application.-Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.-The following test shall be performed by the Government:

4.3.3.6.1 Ballistics.-Twenty-one assembled cartridges from the initial production sample and 21 cartridges from each subsequent production lot shall be grouped and temperature conditioned as prescribed in Table IV and 4.4.6.1.1. Failure of any cartridge to comply with the requirements of 3.4.2 shall be cause for rejection of the lot.

TABLE IV

Temperature	Not Vibrated	Vibrated
-65°F to -70°F	3	4
65°F to 75°F	3	4
160°F to 165°F	3	4

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 11731602.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

4.4 Test methods and procedures.

4.4.1 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.2 Airtightness.-The method for determining airtightness shall be by use of equipment listed on IEL 11731602.

4.4.3 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.4 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case, head and cartridge insert shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.5 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the selaing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional.

4.4.6.1 Ballistics.-The cartridge shall be loaded into an acceptable M2A1 or M3A1 Remover, divided into 3 equal groups and temperature conditioned in accordance with Table I. The cartridge shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the remover shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-A simulated M2A1 or M3A1 remover loaded with a dummy M31A2 cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity: Level A - Drawings C9258146 and C9258147.
Level B - Drawings C9258146 and C9258148.

6. NOTES

6.1 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.-Procurement documents shall specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A578

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. *DoD contractors, government activities,* or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. *If there are additional papers, attach to form and* place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.

6095 SLS

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Philadelphia, Penna. 19137

FOLD

80 6095

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:19Z
Check the source to verify that this is the current version before use.